Final Report: Auditory Stimulation Classification

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https://github.com/fengyiran1005/EEG-Auditory-Stimulation-CNN

Introduction

The human brains exhibit different neural activity patterns sensitive to external stimuli. The project aims to develop a model that would perform a binary classification between EEG of individuals listening to music and individuals in a quiet resting state. The project is motivated by the significance of EEG signals in various neuroscience research. For example, EEG remains the primary source for epilepsy diagnosis and may also serve as a monitor for patients with neurological disorders, brain tumors, brain injury, etc (1). Without deep learning, EEG signals were primarily examined visually for feature extraction, leading to inaccuracies and misinterpretations due to bias or oversight. The primary focus of this project will be a Gated Recurrent Unit (GRU) Recurrent Neural Network, selected for its ability to model temporal dependencies inherent in neural time-series data. Specifically, a 1D Convolutional Neural Network (CNN) will serve as the performance baseline since both models yielded an accuracy above 90% while CNN demonstrated poorer performance on testing data (2).

Background & Related Work

Electrophysiological behavior at auditory stimulation provides meaningful insights into neuronal health. Francavilla et al. discovered that SYNGAP1 haploinsufficiency, a neurological disorder causing neurological disorders like developmental delay, epilepsy, etc, leads to auditory processing deficits (3). Jadhav et al. studied the EEG of SYNGAP1 haploinsufficient mice, confirming that the abnormal oscillations during auditory stimuli are distinguishable with quantitative EEG analysis, demonstrating that neural oscillation patterns serve as biomarkers for auditory processing dysfunction in neurodevelopmental disorders (4). Therefore, a deep learning model in EEG during auditory stimuli would facilitate diagnosis and characterization of such neuronal abnormalities.

Qu et al. used a traditional Random Forest model and Cascade Forward Backpropagation Neural Network (CFBNN) to analyze EEG during neural stimuli to improve disorders of consciousness (DOC) diagnosis. Specifically, they extracted ApEn values from raw EEG data and used a 3 layer CFBNN, which yielded a 83.33% accuracy (5). Aellen et al. adopted the CNN approach to perform a binary prediction of survival and non survival from coma. Specifically, the model used a CNN model with the filter length of the first convolutional layer being 512, then the ReLu activation function, binary cross-entropy loss function and the Adaptive Moment Estimation (Adam) with a learning rate of 5×10^6 was applied (6). Vandecappelle et al. have also used the CNN approach to study EEG during auditory stimulation to identify the auditory attention when multiple speakers are present. The model inputted a 64xT EEG matrix to a single convolution layer consisting of 5 spatio-temporal filters producing 5 feature maps and was followed by a ReLU activation function. The model resulted in a binary classification of attention direction, either left or right (7).

During fine tuning of the GRU model, further research was done to look for reference designs for improving GRU performance. Gao et al. proposed that a parallel feature fusion network combining GRU and CNN yielded an accuracy of 80.7% in classifying 4-class motor image EEG signals (8). This idea was adopted in this project; details can be found in the following sections.

Data Processing

The data was obtained from Physionet (10), where 20 volunteers were recruited, and EEG was collected at 200Hz located at 4 channels for 2 minute intervals. Specifically, each participant was

recorded for 6 sessions in resting state and 6 sessions in auditory stimulation. The resting state was divided into eyes open and eyes closed and the auditory stimulation was classified by different music genres and headphone types. Nevertheless, to train this binary classification model, the dataset will simply be classified into resting and stimulation with no dataset discarded to ensure a large and diverse data population to recognize different types of auditory stimuli. The data provided in 1D time series was segmented and filtered with noise minimized. For this project, each of the 2-minute EEG signals will be z score normalized per channel to preserve characteristics at each electrode. Moreover, the 2-minute interval containing 24000 time points will be segmented into 4 30-second windows with 6000 points to increase the sample count and decrease the training complexity (11). As a result, 240 data sets were generated respectively for resting and music listening, where 80% of the data were used for training, 10% for evaluation and 10% for testing.

Architecture

Including the baseline CNN model, 4 models were implemented. The illustration of the 4 models architectures are shown in figure 1.

Baseline CNN: The baseline CNN consists of three 1D convolutional layers with kernel size 7 and increasing filter depths from 32 to 64 to 128, each followed by ReLU activation and max pooling with factor 4. The resulting feature map is flattened and passed through three fully connected linear layers to produce the final binary classification.

Basic GRU: The basic GRU model involves three stacked GRU stages of hidden size 64, with average pooling of 4 between stages to compress the sequence. The final stage uses a bidirectional GRU whose forward and backward hidden states are concatenated to form a 128 dimensional representation. The representation is then passed through a two-layer classifier for binary classification.

GRU with CNN Front End: The model first applies a CNN compressor of 2 layers that output the same number of channels, so the GRU receives the same feature dimensionality as the raw signal but with shortened sequence. The compressed sequence is then passed through a 2-layer bidirectional GRU with hidden size 32, whose final forward and backward hidden states are concatenated to form a 64-dimensional representation and classified through a two-layer head with dropout of 0.6.

GRU and CNN Parallel Feature Fusion Network: Inspired by Gao et al., this parallel fusion model processes the same 4-channel through two independent branches simultaneously. The CNN branch extracts spectral and spatial features through two 1D convolutional layers followed by adaptive average pooling. The GRU branch extracts the temporal features through two stacked GRU layers with average pooling between them, also producing a hidden size of 32. The two outputs of size 32 are then concatenated into a 64-dimensional representation and classified through a two-layer head.

Baseline Model

CNN was chosen as the baseline due to its steady performance reported in various studies for EEG analysis and classification as mentioned in *Background and Related Work*. Nevertheless, it has been found that it still has a large room for improvement, especially in its performance on testing data (2).

In training the CNN, the set up and data usage for this project were decided. For example, data splitting was changed from 7:1.5:1.5 for training, validation, testing to 8:1:1 since it was found that larger training dataset contributed to a higher accuracy in both validation and testing. The decision of input data normalization was also examined. Specifically, although the 4 electrode channels presented similar mean values, their variance differ slightly. Training with non-normalized dataset demonstrated higher loss and lower accuracy in validation and testing.

Quantitative Result

To quantitatively assess the outcomes of each model, their respective final training, best validation, and test accuracies were compared. Moreover, inspired by Carvajal-Dossman et al. the Area Under the Receiver Operating Characteristic Curve (AUROC) (2), a performance metric for binary classification, was used to quantitatively assess the 4 models in this project. This metric was also included due to the slight imbalance between the control and music group as seen in the confusion matrices in the

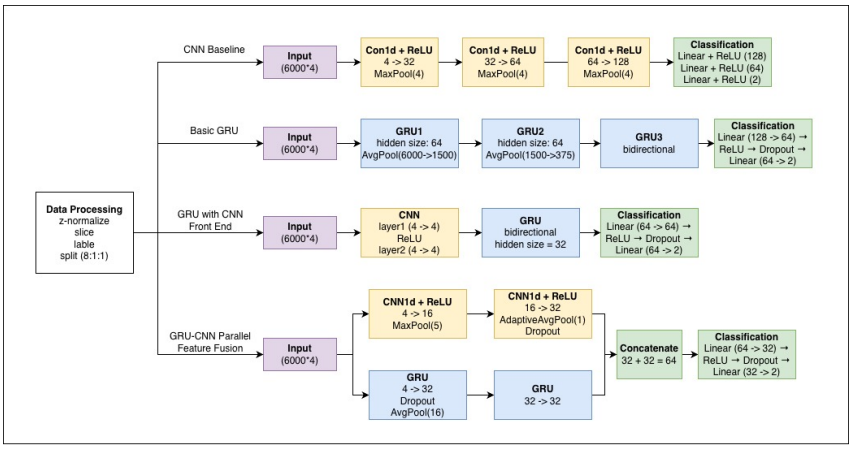

Figure 1: Architecture of 4 models implemented by this project.

following section (Figure 2 to Figure 5). The better the model is at separating the two classes, the closer the AUROC is to 1. The accuracies and AUROC values are shown in Table 1. The training and ROC curves for each model are shown in Appendix A and B.

Model	Training	Validation	Test	AUROC
CNN	80.53%	74.47%	68.42%	0.7429
Basic GRU	78.81%	84.04%	71.58%	0.7529
GRU with CNN front end	79.21%	82.98%	76.84%	0.8062
GRU-CNN parallel feature fusion	88.08%	81.91%	81.05%	0.8743

Table 1: Quantitative comparison of model performances.

Qualitative Results

The goal of this network was to perform a binary classification on the numerical representation of EEG signals where the participant is in silence vs listening to music. The confusion matrix for each model on the testing data was computed and shown in Figure 2 to 5.

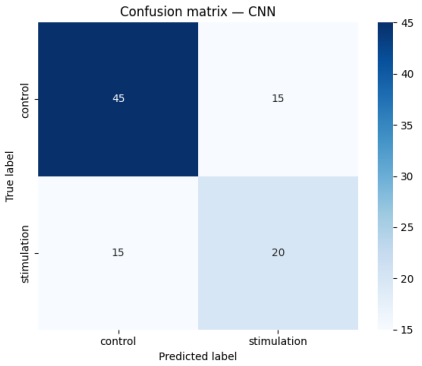

Figure 2: Architecture of CNN model.

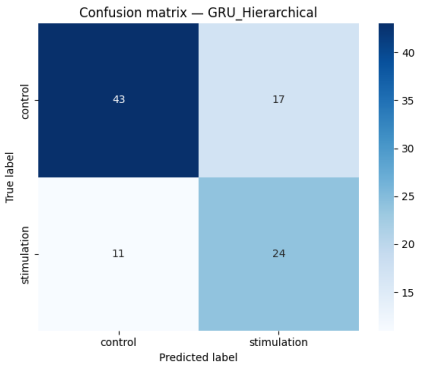

Figure 3: Architecture of hierarchical model.

Evaluate on New Data

To ensure a larger number of training data over validation and testing data, out of the total 944 slices of EEG signals, 95 were reserved for testing. Note that the testing data were data that the network has

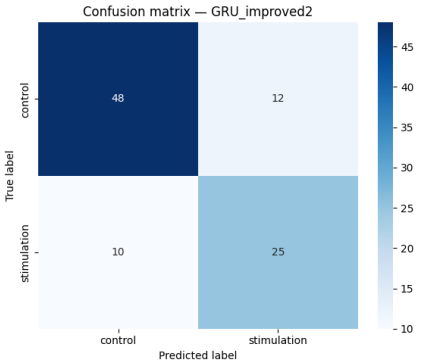

Figure 4: Architecture of GRU with CNN front end model.

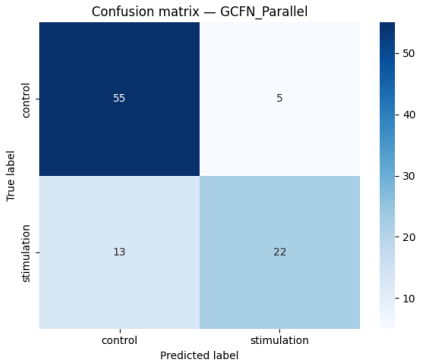

Figure 5: Architecture of GRU CNN fusion network model.

never seen until testing stage. It differs from the validation data, which were used during training for hyperparameter tuning, primarily demonstrating the performance, and learning progress of the model.

The accuracy and confusion matrices are shown respectively in Table 1 and Figure 2 to Figure 5 to demonstrate the models performance on unknown data. Overall, the GRU CNN fusion network achieved a highest testing accuracy of 81.05%, while the GRU with CNN front end reached a 76.84% accuracy and the basic GRU and CNN had accuracies around 70%.

Discussion

As shown in Table 1, the GRU-CNN parallel feature fusion network achieved the highest test accuracy and AUROC value among all models. Specifically, it attained a testing accuracy of 81.05% and an AUROC of 0.8743, largely outperforming the other models. This result was expected, given that the basic GRU model had only shown a slight improvement over the baseline CNN. This observation suggests that EEG signals inherently possess both spatial and temporal features, making the parallel fusion network an optimal choice since it extracts both feature types simultaneously. Furthermore, the parallel fusion network required only 50 epochs to reach optimized loss and accuracy, whereas the other GRU models needed 100 epochs.

From the confusion matrix, no significant biases were observed due to the small dataset size. In future works, more data can be collected to improve variability of data used in training to achieve better performances. The need for a larger dataset is also confirmed by many studies reporting accuracies of EEG classification over 90% using purely CNN or GRU (2). This finding indicates that 70% to 80% of accuracy still has large room for improvement before reaching the model's maximum capacity.

Other reasons that could have possibly contributed to the accuracy remaining at 70% to 80% include non-strictly controlled experiment setting and individual differences, which have introduced generalization in the training but also complexity. Specifically, the control group had two states of eye closed and eye open; the auditory stimulated group data collection also varied in language of song listened to, type of headphone, and music type. Moreover, since EEG primarily records brain activity, the recordings were susceptive to body movement, eye blink, mood swing, stress, etc.

Overall, the model performance was satisfactory given the short time constraints and limited access to data. Future improvement can be made with larger dataset, especially more testing dataset, to assess bias and insufficiencies in the model.

Ethical Considerations

The data from Physionet (10) was collected with informed consent from all participants with their identity masked. Throughout this project, there will be no attempt to re-identify the participants or misuse the data. More rigorous tests of this model should be performed if being applied in health care to avoid inaccurate predictions. Furthermore, due to the small number of participants, the data might not have a full representation of the population, potentially leading to bias in the algorithm.

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A Training and Validation Loss and Accuracy Curves

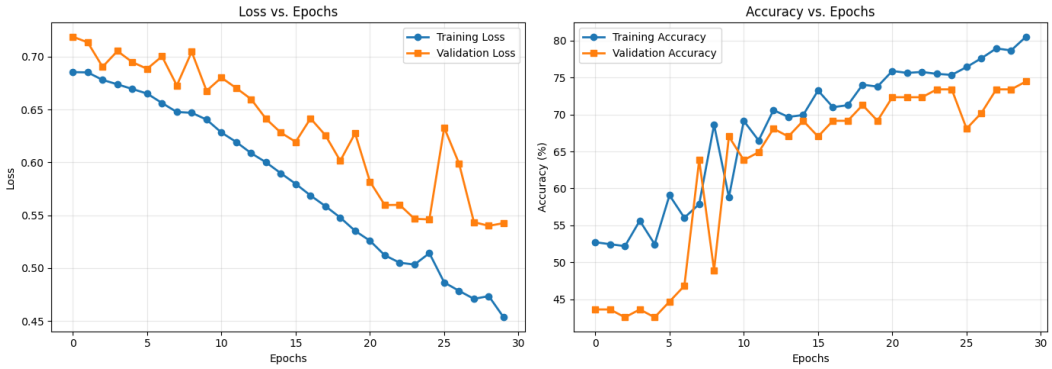

Figure 6: Baseline CNN training curves.

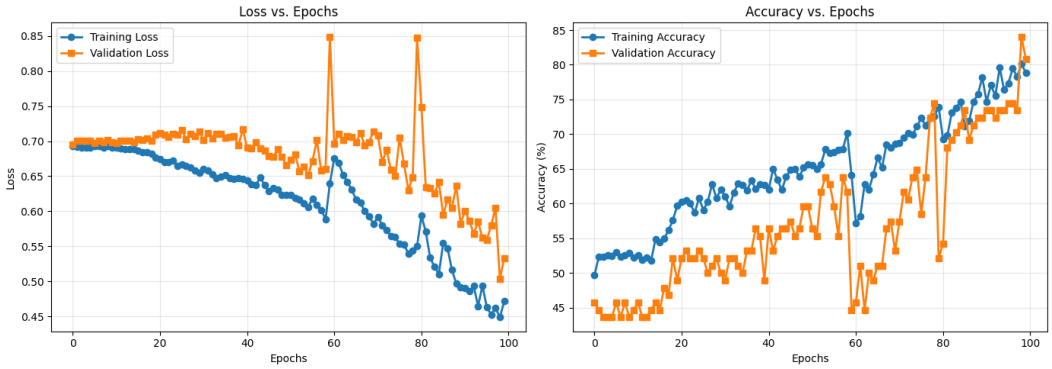

Figure 7: Basic GRU training curves.

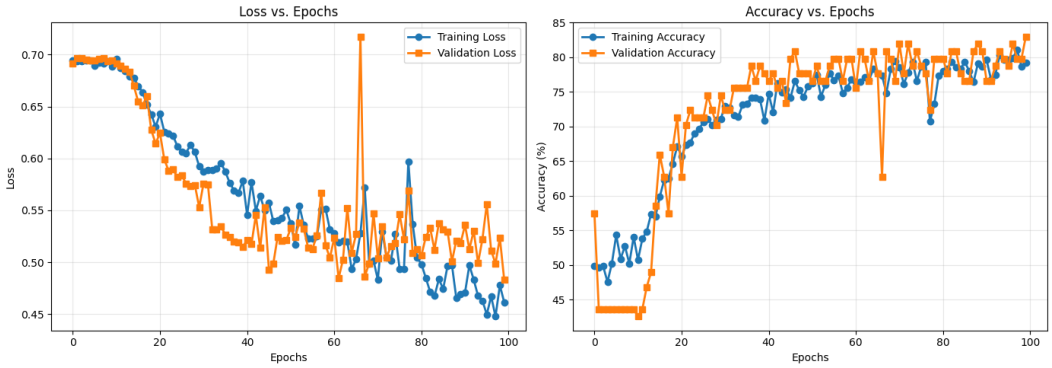

Figure 8: GRU with CNN front end training curves.

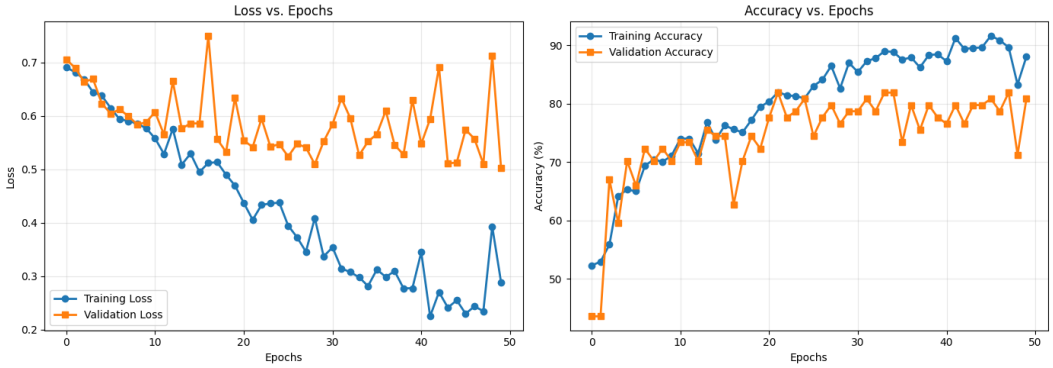

Figure 9: GRU CNN parallel feature fusion training curves.

B ROC Curves

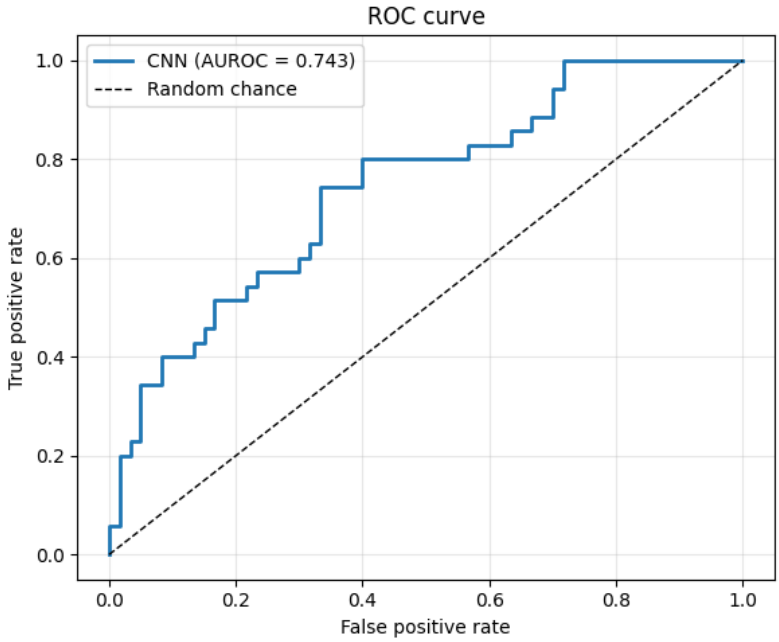

Figure 10: Baseline CNN ROC curve.

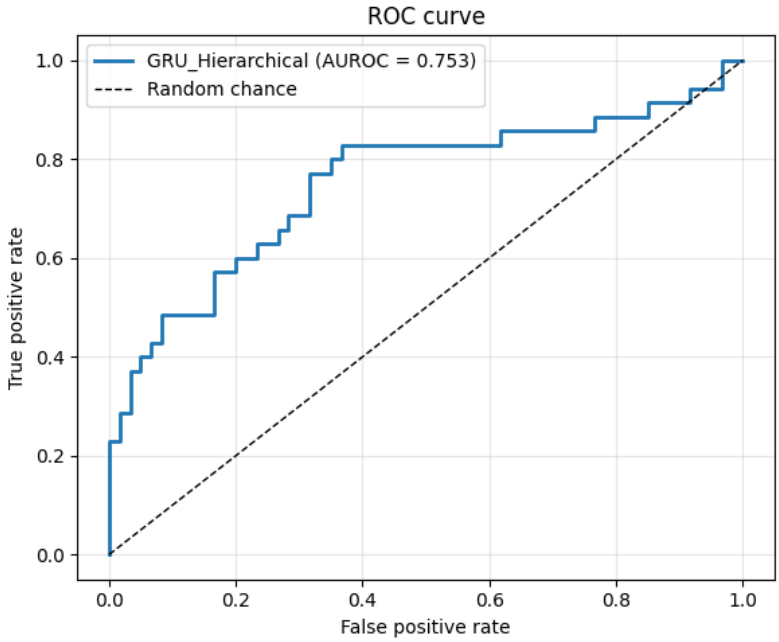

Figure 11: Basic GRU ROC curve.

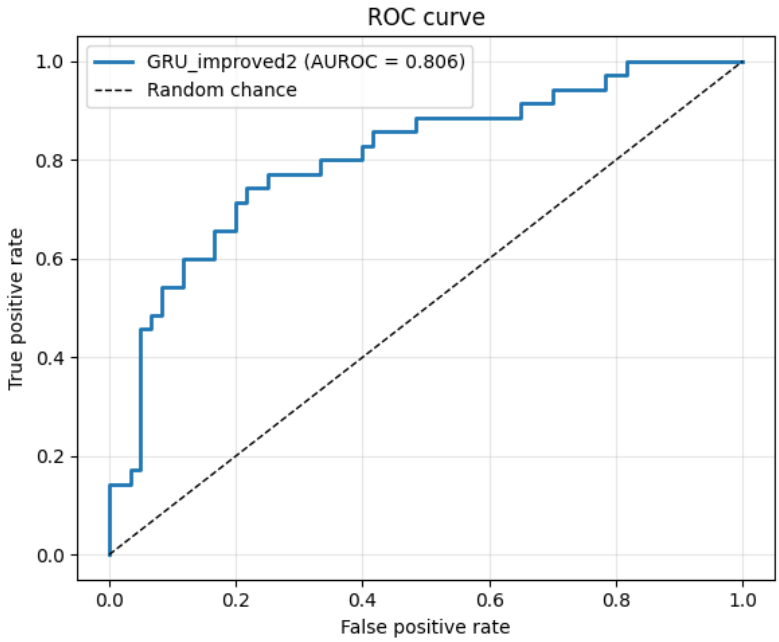

Figure 12: Basic GRU ROC curve.

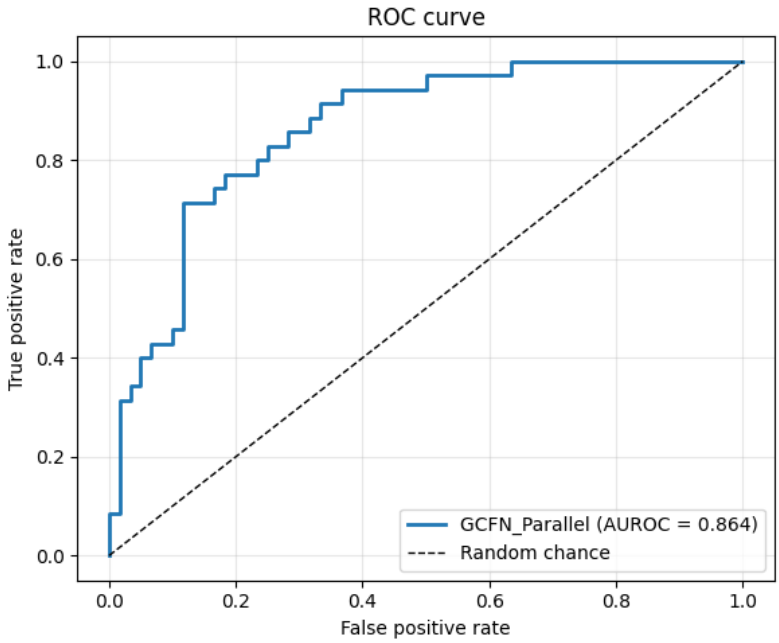

Figure 13: Basic GRU ROC curve.